Retrieval Survives, Choice Collapses: Induction Heads under Recursive Synthetic Training

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Abstract

We train two-layer Transformers on the in-context symbol–value task of Singh et al. and retrain each generation on labels sampled from its predecessor. A pilot with a short training budget never left the pre-induction plateau, so we first make induction heads form at generation 0 (100% accuracy, causal head-ablation drops up to 70 points). Predicting one value shows no collapse. When the model must also choose a next symbol and give its value, consistency falls linearly (99.4%→94.8% over five generations) as probability leaks to symbols absent from the context (0.7%→4.9%). Retrieval is untouched: values of shown symbols stay ≥99.6% correct and induction attention stays at ceiling. Direct logit attribution traces the leak to a shrinking preference for in-context symbols at the choice step, 78% of it in the layer-2 heads. Keeping half of every generation's labels real stops the drift at a ceiling predicted by a simple error-propagation model.

1 Introduction

In-context learning (ICL) lets a Transformer infer a mapping from its prompt without weight updates, and induction heads are a well-understood mechanism for it [2, 5]. Model collapse is the degradation of models trained on earlier models' outputs [3, 4]. We ask whether recursive training damages the ICL *mechanism* or only what the model chooses to produce. On a task with a known circuit we show that (i) a short training budget yields a non-induction heuristic that must be ruled out first; (ii) a fully context-determined target does not collapse; (iii) when the target includes a free choice, collapse appears as a steadily weakening preference for in-context symbols, located mostly in layer-2 heads, while retrieval stays intact; and (iv) mixing in real data bounds the damage, as a simple error-propagation model predicts.

2 Background

In a two-layer induction circuit a layer-1 *previous-token head* copies each token's predecessor into its residual stream, and a layer-2 *induction head* matches the current token against those keys and copies the token that followed [2]. On this symbol–value task the circuit forms in an abrupt phase transition after a long loss plateau [5]. Recursive training compounds sampling and approximation error, so learned distributions lose their tails and drift [3, 4]. Circuit claims should rest on causal interventions, not attention maps alone [1].

3 Method

Task. Each sequence draws a fresh bijection between 8 symbol and 8 value tokens. The context is `BOS`, six symbol–value pairs (symbols with replacement) and a query symbol shown earlier. The *base* target is the query's value. The *extended* target adds a next symbol, drawn uniformly from the distinct shown symbols, and its value. Our pilot drew it from all 8 symbols, making its value unknowable in 45% of cases. The redesign keeps every target inferable and gives the symbol choice a known uniform reference.

Model and training. A 2-layer causal Transformer (4 heads, width 64, MLP 128, pre-LayerNorm, learned positions, tied embeddings) is trained with AdamW (weight decay 10^{-4}, batch 128) on cross-entropy over target tokens. We use 50k/2k/2k train/validation/test sequences and keep the checkpoint with the lowest validation loss (max. 20 epochs, patience 6). Learning rate $\in \{3{\cdot}10^{-4}, 10^{-3}, 3{\cdot}10^{-3}\}$ and training size $\in \{3\text{k}, 50\text{k}\}$ are selected on validation data only.

Recursion. Generation 0 trains on real labels. Generation $g{\geq}1$ is a freshly initialised model trained on the same contexts, where a fraction f of labels is sampled at $T{=}1$ from generation $g{-}1$ and the rest stay real. We run $f{=}1$ for both tasks and $f{=}0.5$ for the extended task, each for 6 generations $\times$ 3 seeds. Validation and test labels are always real.

Metrics. *Pair consistency* is the fraction of sampled extended outputs whose symbol is shown and whose value matches it. Exact match is capped near 0.24 because the next symbol is random by design. The *induction score* is the attention from the query to values following earlier occurrences of the query symbol (uniform attention over values scores 0.27). Head *ablation* zeroes one head and records the drop in first-value accuracy. *Leaked mass* is the probability placed on unshown symbols at the choice step. For *direct logit attribution* (DLA) we freeze the final LayerNorm scale and split the choice-step logits exactly into contributions from embeddings, each head and each MLP. For each component we report the mean logit it gives shown minus unshown symbols.

4 Results

Pilot and tuning. With 3k examples (≈600 steps) generation-0 accuracy was $32.0 \pm 3.1\%$, between the heuristics "copy a random context value" (25.8%) and "copy the most frequent value" (36.8%), with induction scores (0.24) at the uniform baseline. Every learning rate stayed at 30–34% with 3k examples. With 50k, 10^{-3} and $3{\cdot}10^{-4}$ reached 100% and $3{\cdot}10^{-3}$ stalled at 70%. We chose 10^{-3}, which crossed 99% fastest (1,955 steps; Fig. 1a).

Base task: no collapse. All 18 base models score 100% at every generation, their synthetic labels are ≥99.998% correct, and induction scores stay at 0.90–1.00.

Extended task: choice collapses, retrieval does not. First-value accuracy stays at 100%, but pair consistency falls by 0.9 points per generation (Fig. 1b, Table 1). Nearly every failure is an unshown symbol. When the symbol is shown its value is

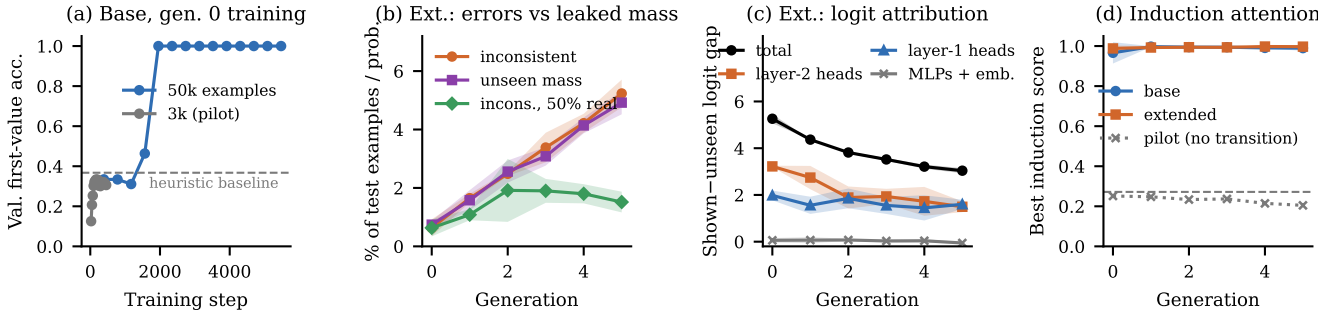


Figure 1: (a) Generation-0 validation accuracy: with 3k examples training stops on the plateau, while with 50k the induction transition occurs at ≈1,600–2,000 steps. (b) Extended task, f=1: inconsistent pairs and mass on unshown symbols grow together; with 50% real labels (green) errors level off. (c) DLA at the choice step (f=1): the shown−unseen logit gap shrinks, mostly in layer-2 heads. (d) Induction attention stays at ceiling; the pilot stayed at the uniform baseline (dashed). Bands: ±1 SD over 3 seeds.

Table 1: Clean-test results (mean ± SD over 3 seeds). Extended rows use f=1 unless marked.

Task	Metric	Gen. 0	Gen. 5 (f=1)	Gen. 5 (f=0.5)
Base	First-value accuracy	1.000 ± 0.000	1.000 ± 0.000	–
Extended	First-value accuracy	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
Extended	Sampled pair consistency	0.994 ± 0.003	0.948 ± 0.005	0.985 ± 0.004
Extended	Leaked mass on unshown symbols	0.007 ± 0.001	0.049 ± 0.004	0.017 ± 0.002
Extended	Unshown symbols in training labels	0.000 ± 0.000	0.041 ± 0.002	0.009 ± 0.001
Extended	Shown−unshown logit gap (DLA)	5.26 ± 0.19	3.04 ± 0.11	4.26 ± 0.16
Extended	Best induction score	0.988 ± 0.015	0.997 ± 0.002	0.994 ± 0.006

correct in ≥99.6% of cases, and greedy decoding stays 100% consistent. Leaked mass tracks the errors (0.7%→4.9%), and the training labels skew: the top symbol's share rises from 12.7% to 17.5%.

Mechanism. Induction scores (0.97–1.00) and causal ablations (largest single-head drop 9–70 points in every model) show an intact retrieval circuit. DLA shows what changes instead: the total logit gap favouring shown symbols shrinks from 5.26 ± 0.19 to 3.04 ± 0.11, monotonically in every seed. The layer-2 heads account for 78% of this ($3.22 \rightarrow 1.49$), layer-1 heads change less and less consistently ($1.98 \rightarrow 1.60$), and MLPs and embeddings contribute almost nothing (Fig. 1c). No component's symbol-specific bias grows, so the skew reflects leaked mass flowing to pre-existing preferences rather than a new symbol prior.

Mixing real data. With f=0.5, consistency settles at 98.1–98.5% from generation 2 onwards, leaked mass at 1.7–1.8% and the logit gap at 4.2–4.3. Generation 0 is identical in both runs, confirming the comparison is controlled.

5 Discussion and limitations

A context-determined lookup is a fixed point of recursion: a perfect model emits perfect labels. A uniform choice is not. Each model puts a little mass, δ, on unshown symbols, sampling turns it into labels, and the next model learns those labels as legitimate data. This coincides with a shrinking shown-symbol preference in the layer-2 heads, and the unshown-label rate follows $L_{g+1} = f(L_g + \delta)$. With $\delta \approx 0.76\%$ (the generation-1 label rate at f=1), this predicts linear growth $L_5 = 5\delta \approx 3.8\%$ for f=1 (observed 4.1%) and a ceiling $f\delta/(1-f) = \delta \approx 0.76\%$ for f=0.5 (observed 0.9%). Unlike the tail loss of [4], the damage here is *spurious* tail mass that grows while the mode, greedy decoding and the induction circuit stay perfect. The circuit does not break. What erodes is how strongly the layer-2 heads restrict the free choice to context symbols, and real data caps that erosion. Collapse in ICL can therefore hide behind intact circuits and perfect accuracy, and only distributional metrics on sampled outputs expose it.

Limitations. Only labels are recursive (contexts stay real), at T=1. Five generations, three seeds, one mixing level, 8 symbols and 2 layers limit generality. Early stopping on real validation data gives each generation some real signal. DLA freezes the LayerNorm scale and attributes logits, not the attention weights behind them. Ablation of single heads understates redundancy. Future work: path patching to see which attention edges carry the weakened preference, and more values of f and T.

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Contribution statement

This is an individual submission. Sibonelo Mlambo (536187) carried out the experimental design, implementation, experiments, analysis, figures and writing.

Code and reproducibility

Code, configurations, raw results and checkpoints accompany the submission. `tune.py` reproduces the hyper-parameter selection, `run_study.py` the main study, `analyse_leakage.py` and `analyse_mechanism.py` the leaked-mass and attribution analyses, `configs/mix50.yaml` the mixing run, and `make_figures.py` Figure 1. All experiments ran on CPU without a GPU. The main study (3 seeds $\times$ 2 tasks $\times$ 6 generations, 36 training runs on 50k sequences) takes about 2–2.5 hours on a 2-core CPU. The other reported experiments add 18 runs for the mixing ablation (extended task only), 6 runs for the tuning grid (three of them on 3k sequences) and 36 runs for the pilot on 3k sequences.

Broader impact

Recursive training on generated data is a growing concern as model outputs enter training corpora. Our results suggest a practical lesson: collapse can grow while accuracy, greedy outputs and the underlying circuit all look intact, so monitoring should include distributional metrics on sampled outputs, and retaining real data limits the damage. The models are tiny and trained on synthetic tokens, so we see no direct path to misuse; the main risk is over-generalising these small-scale findings to production language models, which the limitations above caution against.

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    Answer: [NA]

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AI ethics statement

ACTION REQUIRED BEFORE SUBMISSION. Replace this box with the official Faculty of Science AI Ethics Statement by saving it as `report/ai_ethics.tex`.

Declaration of AI use (draft). Generative AI (Claude, Anthropic) assisted with code development and debugging, running and analysing experiments, producing figures, and drafting and editing this report. All experiments were run on the submitted code, and every reported number comes from the saved result files. I have reviewed the code and text, can explain them, and take full responsibility for the correctness and integrity of this submission.